Zero Classification and Exact Finite Minima for OEIS A085124

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Abstract

We prove that a positive integer has a multiple whose decimal digit product is a positive multiple of that integer if and only if it is 7-smooth and is not divisible by 10. This settles the existence conjecture recorded in OEIS A085124. A finite automaton gives the exact minimum at every index, with a proved bound on its number of digits. The minimum formula retains finite minimization.

1 An elementary construction

Let $P(k)$ be the product of the decimal digits of a positive integer k. OEIS A085124 [1] defines

$$a(n) = \min\{k \geq 1 : n \mid k, \quad 0 < P(k), \quad n \mid P(k)\},$$

with zero when the set is empty.

Lemma 1. *For $p \in \{2, 5\}$ and $e \geq 1$, there is a zero-free decimal word of length e whose value is divisible by p^e.*

Proof. Start with the empty suffix of value zero. Given a suffix of length $j-1$ and value s divisible by p^{j-1}, select its new leftmost digit $t \in \{1, \ldots, p\}$ so that

$$s + t10^{j-1} \equiv 0 \pmod{p^j}.$$

After division by p^{j-1} the coefficient of t is coprime to p, so exactly one of these p nonzero digits works. Induction proves the assertion. □

Theorem 1. *For every $n \geq 1$, $a(n) > 0$ if and only if n is 7-smooth and $10 \nmid n$.*

Proof. A positive digit product has no prime factor greater than 7. A multiple of 10 ends in zero and has zero digit product. These prove necessity.

Conversely write $n = 2^a 3^b 5^c 7^d$ with $\min(a, c) = 0$. Put $C = 2^a 5^c$ and $m = n/C$, so $\gcd(m, 10) = 1$. If $C > 1$, apply the lemma to its unique prime base to obtain a zero-free suffix T of length $a + c$ divisible by C. Otherwise put $T = 1$. Let W consist of a digits 2, b digits 3, c digits 5, and d digits 7, and concatenate $B = WT$. The block B is nonempty and zero-free, $n \mid P(B)$, and $C \mid B$, since both T and $10^{|T|}$ are divisible by C.

Put $L = |B|$ and $h = \mathrm{ord}_m(10^L)$ if $m > 1$, or $h = 1$ if $m = 1$. Repeating B exactly $R = mh$ times yields

$$K = B \sum_{j=0}^{R-1} 10^{Lj}.$$

This is divisible by C. Modulo m, the geometric sum consists of m equal groups of length h, so it is divisible by m. Finally $P(K) = P(B)^R$ is a positive multiple of n. Thus K is a witness. □

2 The all-index minimum

For an admissible n, let $e = (a, b, c, d)$ and index its coordinates by $p \in \{2, 3, 5, 7\}$. Use ordinary automaton states (r, f), where $0 \le r < n$ and $0 \le f_p \le e_p$, together with an extra initial vertex. Reading a digit $t \in \{1, \dots, 9\}$ gives

$$(r, f) \longmapsto ((10r + t) \bmod n,\ (\min(e_p, f_p + v_p(t)))_p)\,.$$

The extra initial vertex has the outgoing transitions of $(0, 0)$ but is never accepting. The only accepting ordinary state is $(0, e)$.

Theorem 2. *Let $N = n(a + 1)(b + 1)(c + 1)(d + 1)$. The value $a(n)$ is the integer represented by the shortest accepted word, with ties broken lexicographically. Its length is at most N. Breadth-first search with digits in ascending order computes it exactly.*

Proof. The residue coordinate enforces divisibility of the represented integer; the other coordinates enforce divisibility of its positive digit product. The initial vertex enforces a nonempty word, including at $n = 1$. Thus accepted words are exactly the required decimal expansions. They are nonempty by the construction above. Among such expansions length followed by lexicographic order is precisely numerical order.

If a shortest accepting path repeated an ordinary state, deleting the intervening cycle would yield a shorter nonempty accepting path. There are N ordinary states, proving the digit bound and hence finiteness of the formula. □

3 Reproducible checks and scope

The accompanying standard-library Python program compares the first 50 displayed OEIS terms, checks 51 independent minima from a direct integer scan through 200000 for indices at most 200, and verifies all 73 admissible constructive witnesses through index 500. These finite checks validate the implementation; the theorems establish the infinite assertions. The exact formula uses finite-state minimization and is not an elementary expression without it. No claim of an OEIS submission is made.

References

[1] OEIS Foundation Inc., *A085124*, https://oeis.org/A085124, accessed October 26, 2026.